

NHS ERIC 2024/25 — Business Analysis & Power BI Case Study

Executive summary

Independent BA case study using publicly available NHS England ERIC 2024/25 site-level data. The objective was to demonstrate an end-to-end approach to a business-analysis scenario: problem definition, data investigation, findings, requirements, process improvement, solution design, testing and expected value.

Important scope statement

This is an independent portfolio scenario, not an NHS implementation and not evidence of NHS stakeholder engagement.

Business problem

How could potentially unusual estate-finance observations be identified and investigated more consistently? The project deliberately avoids treating an unusual observation as proof of poor performance.

Data investigation

- 2,901 site records analysed.
- 2,901 unique Site Codes.
- 28 blank Occupied Floor Area records; all 28 were Unoccupied sites.
- 0 cases where Occupied Floor Area exceeded Gross Internal Floor Area.
- 2,873 valid finance-cost-per-occupied-m² observations.
- 1 negative calculated finance-cost-per-occupied-m² observation requiring contextual investigation.

Findings

FACT: the calculated finance-cost-per-occupied-m² measure varies substantially. INTERPRETATION: some observations may warrant contextual investigation. LIMITATION: ERIC alone does not establish cause or poor performance. RECOMMENDATION: use analytical flags as prompts for human investigation.

Requirements

- Identify site records and calculate the analytical measure.
- Apply validation rules.
- Screen potentially unusual observations.
- Present relevant context.
- Explain why a record was flagged.
- Preserve original source values.
- Provide traceability and explainability.

AS-IS / TO-BE

AS-IS: access data → review measures → identify unusual observation → check definition/context → decide whether investigation is required. TO-BE: prepare/refresh → calculate → validate → screen → present context → human investigation → close or refer.

Solution

Five options were considered. The recommended MVP is Power BI because the validated requirements were primarily analytical: calculation, validation, filtering, exception screening, drill-down and contextual investigation. Power Apps, Dataverse and Power Automate were kept out of scope because no validated requirement for operational case management or workflow was established.

Testing

Technical/data tests were executed against the underlying dataset. Power BI UAT scenarios were designed, but full Power BI Desktop UAT and production deployment were not completed.

Expected benefits

- Faster identification of potential exceptions.
- More consistent review using documented rules.
- Better context at the point of investigation.
- Improved transparency through explainable flag reasons.
- Better visibility of data-quality issues.

Key lesson

The system should handle repeatable calculation, validation and screening; human users should retain responsibility for contextual interpretation and decisions.